

The Triadic Cycle as Unified Framework

A Foundational Ontology of Information, Energy, and Geometry

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Abstract

We introduce the *Triadic Cycle*, a minimal and closed ontological structure consisting of three fundamental roles—information, energy, and geometry—and show that modern physics already implements this cycle across its most established theories. Rather than proposing new physical laws, the Triadic Cycle is presented as an architectural extraction: a unifying ontology implicit in quantum mechanics, general relativity, thermodynamics, quantum information theory, and holography. We demonstrate how each domain instantiates one or more triadic transitions, forming the closed sequence $E \rightarrow G \rightarrow I \rightarrow E$.

Black holes serve as the extreme and most transparent realization of the cycle, where energy determines geometry, geometry bounds information, and information constrains energetic evolution. This regime also reveals that time is not fundamental but emerges from triadic interactions between change, causal structure, and distinguishability. Conservation of energy, unitarity, and causal preservation are shown to correspond directly to the conservation of the triadic roles themselves.

We further argue that the concepts of dark matter and dark energy are ontologically misclassified within the traditional “matter–energy” vocabulary. Their observed behaviour is incompatible with the energetic or informational roles of the Triad and is more coherently interpreted as structural geometric constraints rather than physical substances.

Finally, we show that mainstream theories can be understood as projections of the Triadic Cycle onto single roles: quantum mechanics as an information-first projection, quantum field theory as an energy-first projection, and general relativity as a geometry-first projection. This architectural perspective does not replace existing theories but clarifies their interrelations and resolves several conceptual tensions. The Triadic Cycle thus offers an ontological structure that future unification efforts or theories of everything can build upon and remain consistent with.

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1 Introduction

Modern physics exhibits an extraordinary degree of empirical success while remaining conceptually fragmented. Quantum mechanics, quantum field theory, general relativity, thermodynamics, and holography each describe essential aspects of physical reality, yet they do so using distinct ontological primitives and incompatible structural assumptions. Attempts at unification typically introduce new mathematical frameworks or speculative microphysical models, but these efforts rarely address the deeper question of whether a minimal and coherent ontological foundation underlies the established theories themselves.

This paper proposes that such a foundation already exists implicitly within mainstream physics. We introduce the *Triadic Cycle*, a minimal ontological structure consisting of three fundamental roles—information, energy, and geometry—connected through a closed sequence of transformations. Rather than functioning as a new physical theory, the Triadic Cycle is presented as an architectural extraction: a structural grammar that is already instantiated across the major theoretical frameworks of contemporary physics. Each domain emphasizes one or more triadic roles, and the tensions between theories can be understood as tensions between their respective projections of the same underlying cycle.

Black holes provide the clearest and most extreme realization of this architecture. In these systems, energy determines geometry, geometry bounds information, and information constrains energetic evolution. The black-hole regime also reveals that time is not a fundamental parameter but an emergent phenomenon arising from triadic interactions between change, causal structure, and informational distinguishability. Conservation of energy, unitarity, and causal preservation further correspond to the conservation of the triadic roles themselves, suggesting that the Triad is not an imposed abstraction but a structural feature of physical law.

The Triadic Cycle also offers a conceptual clarification of two of the most puzzling components of modern cosmology: dark matter and dark energy. Their observed behaviour does not align with the energetic or informational roles of the Triad and is more coherently interpreted as structural geometric constraints rather than physical substances. This reframing does not propose new physics but highlights a potential misclassification in the traditional matter–energy vocabulary.

Finally, we show that established theories can be understood as partial or role-specific projections of the Triadic Cycle: quantum mechanics as an information-first projection, quantum field theory as an energy-first projection, and general relativity as a geometry-first projection. This architectural perspective does not replace existing theories but clarifies their interrelations and resolves several conceptual tensions. The Triadic Cycle thus provides a minimal ontological framework that future unification efforts can build upon and remain consistent with.

2 The Triadic Ontology

The Triadic Cycle is built on three fundamental roles—information, energy, and geometry—that appear across all major physical theories. These roles are not introduced as new physical entities or metaphysical primitives; rather, they are extracted from the structural commitments already present in contemporary physics. Each role captures a distinct aspect of physical description, and together they form a minimal closed ontology capable of supporting the full range of known physical phenomena.

2.1 Information: Distinction and State

Information refers to the capacity of a system to instantiate distinguishable states. In quantum mechanics, this role is formalized through Hilbert spaces, entanglement structure, and the unitarity of time evolution Nielsen and Chuang 2010. In thermodynamics and statistical mechanics, information appears through entropy, coarse-graining, and the counting of microstates

Jaynes 1957. In holography, information is encoded geometrically on boundary surfaces Ryu and Takayanagi 2006.

Within the Triadic Cycle, information is not merely a measure but a structural role: it determines what can be distinguished, stored, or transformed. Information constrains physical evolution by limiting the possible transitions between states, and it participates in the cycle by feeding back into energetic and geometric behaviour.

2.2 Energy: Transformation and Dynamics

Energy governs the capacity for change. In classical and quantum physics, energy generates time evolution through Hamiltonian or unitary dynamics. In quantum field theory, energy density determines excitations of fields and the propagation of interactions. In thermodynamics, energy exchange underlies the laws of heat, work, and entropy production.

In the Triadic Cycle, energy is the role responsible for transformation: it drives the evolution of informational states and shapes the geometry through stress–energy contributions. Conservation of energy, arising from time-translation symmetry via Noether’s theorem **noether1918**, reflects the stability of this role across physical theories.

2.3 Geometry: Relation and Causality

Geometry encodes the relational and causal structure of physical systems. In general relativity, geometry is dynamical and determined by the distribution of energy and momentum Einstein 1922. In quantum gravity and holography, geometry emerges from entanglement patterns or tensor-network structures Swingle 2012. Causal structure, light cones, and metric relations define which transformations and informational exchanges are physically possible.

Within the Triadic Cycle, geometry is the role that constrains interaction: it determines locality, causal order, and the relational scaffolding on which information and energy operate. Conservation of causal structure—the preservation of light cones and metric relations—reflects the stability of this role.

2.4 Minimality of the Three Roles

The Triad is proposed as a minimal ontology: removing any of the three roles leads to an incomplete or inconsistent description of physical reality.

- Removing information eliminates the capacity to distinguish states, undermining quantum mechanics, thermodynamics, and holography.
- Removing energy eliminates dynamics, rendering time evolution and physical change undefined.
- Removing geometry eliminates causal structure and locality, making interaction and measurement impossible.

No additional roles appear necessary for the structural coherence of known physics. The Triad is therefore not an expansion of ontology but a reduction to the minimal set of roles that all established theories implicitly rely upon.

2.5 Canonical Physical Relations Underlying the Triadic Cycle

The Triadic Cycle is grounded in four well-established physical relations that connect energy, geometry, and information. These relations are not speculative: they arise directly from relativity, gravitation, thermodynamics, and information theory. Together, they form a closed sequence of mutual determination.

Energy to Geometry. Einstein’s mass–energy equivalence,

$$E = mc^2,$$

establishes that mass is a concentrated form of energy. In general relativity, this energetic content determines spacetime curvature. For a spherically symmetric mass M , the Schwarzschild radius

$$r_s = \frac{2GM}{c^2}$$

quantifies the geometric deformation produced by that energy. Geometry is therefore not primitive but shaped by energetic content.

Geometry to Information. The Bekenstein bound relates the maximum information I that can be stored within a region of radius R and energy E :

$$I \leq \frac{2\pi ER}{\hbar c \ln 2}.$$

In black-hole thermodynamics, this becomes sharper: the Bekenstein–Hawking entropy,

$$S = \frac{k_B c^3 A}{4G\hbar},$$

shows that information capacity scales with surface area rather than volume. Geometry therefore bounds informational structure.

Information to Energy. Landauer’s principle connects information to energetic cost. Erasing one bit of information requires

$$E_{\text{erase}} \geq k_B T \ln 2.$$

Information is thus physically grounded: its manipulation requires energy, and informational constraints shape energetic evolution.

The Closed Cycle. Taken together, these relations form the minimal closed sequence

$$E \rightarrow G \rightarrow I \rightarrow E.$$

Energy determines geometry; geometry bounds information; information processing requires energy. No additional ontological categories are needed, and removing any link breaks the structure.

Minimality and Recursion. The cycle is minimal because mass is not a separate primitive (being equivalent to energy), geometry is derived rather than fundamental, and information is physically instantiated rather than abstract. The cycle is recursive because each role generates the next: energetic configurations shape geometry, geometric constraints shape informational capacity, and informational processes alter energetic states. This recursive structure appears across scales—from black holes to quantum information to thermodynamic computation—suggesting that the triad is a structural feature of physical law rather than a modelling artefact.

2.6 The Closed Triadic Cycle

The three roles form a closed sequence of mutual determination:

$$E \rightarrow G \rightarrow I \rightarrow E.$$

Energy shapes geometry through stress–energy relations. Geometry bounds information through causal and entropic constraints. Information constrains energetic evolution through unitarity, entropy bounds, and quantum speed limits.

This closed cycle is not imposed but observed across physical theories. It provides the architectural backbone for the remainder of the paper, where we show how established physics instantiates each transition and how extreme systems such as black holes make the full cycle explicit.

3 Established Physics Already Contains the Triadic Cycle

The Triadic Cycle is not introduced as a speculative extension of physical theory. Instead, it is extracted from structural patterns that are already present across the most successful frameworks of modern physics. Each major domain instantiates one or more transitions of the cycle, and in several cases the full sequence is implicitly realized. This section surveys how quantum mechanics, quantum field theory, general relativity, thermodynamics, and holography each embody the triadic roles and their interrelations.

3.1 Quantum Mechanics: Information Constrains Energetic Evolution

Quantum mechanics is fundamentally an information-first theory. States are represented by vectors in a Hilbert space, and physical evolution is governed by unitary transformations that preserve informational distinguishability Nielsen and Chuang 2010. The Hamiltonian generates time evolution, linking information and energy through the Schrödinger equation:

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = H |\psi(t)\rangle.$$

Quantum speed limits, such as the Mandelstam–Tamm and Margolus–Levitin bounds Margolus and Levitin 1998, explicitly connect the rate of informational change to energetic resources. Thus, quantum mechanics realizes the transition:

$$I \rightarrow E,$$

where informational structure constrains energetic evolution.

3.2 Quantum Field Theory: Energy Shapes Interaction Structure

Quantum field theory (QFT) emphasizes the energetic role. Fields carry excitations whose energies determine interaction strengths, propagation, and particle content. Renormalization flows, vacuum structure, and symmetry breaking all depend on energetic configurations Peskin and Schroeder 1995. Locality and causal structure are encoded through commutation relations that vanish outside the light cone, linking energy to geometry.

Thus, QFT realizes the transition:

$$E \rightarrow G,$$

where energetic distributions determine the effective geometric and causal structure of interactions.

3.3 General Relativity: Geometry Determines Information

General relativity (GR) is a geometry-first theory. The Einstein field equations,

$$G_{\mu\nu} = 8\pi G T_{\mu\nu},$$

relate curvature to the stress–energy tensor Einstein 1922. Geometry determines causal structure, light cones, and the set of physically accessible states. In curved spacetimes, the amount of information that can be stored or transmitted is bounded by geometric constraints such as horizon areas and causal diamonds **bousso bound**.

Thus, GR realizes the transition:

$$G \rightarrow I,$$

where geometry constrains informational capacity.

3.4 Thermodynamics and Statistical Mechanics: Information Determines Energy

Thermodynamics links information and energy through entropy, free energy, and the second law. Landauer’s principle Landauer 1961 states that erasing one bit of information requires a minimum energetic cost of $k_B T \ln 2$. Statistical mechanics interprets entropy as the logarithm of the number of microstates, making information the foundation of thermodynamic behaviour Jaynes 1957.

This realizes the transition:

$$I \rightarrow E,$$

completing one half of the triadic cycle.

3.5 Holography: Geometry Encodes Information

The holographic principle ’t Hooft 1993; Susskind 1995 and the AdS/CFT correspondence Maldacena 1998 provide one of the clearest demonstrations of the triadic structure. The Ryu–Takayanagi formula Ryu and Takayanagi 2006 relates entanglement entropy to minimal surfaces in the bulk geometry:

$$S_A = \frac{\text{Area}(\gamma_A)}{4G_N}.$$

Here, geometry directly encodes information, realizing:

$$G \rightarrow I.$$

Tensor-network models of holography further show how entanglement patterns generate emergent geometry Swingle 2012, reinforcing the triadic connection.

3.6 The Full Cycle Across Theories

Taken together, these domains instantiate the entire triadic sequence:

$$E \rightarrow G \rightarrow I \rightarrow E.$$

- QFT shows how energy shapes geometry.
- GR shows how geometry bounds information.
- QM and thermodynamics show how information constrains energetic evolution.

The Triadic Cycle is therefore not an imposed abstraction but a structural pattern already present in the foundations of physics. The remainder of the paper examines how this cycle becomes explicit in extreme systems such as black holes and how it clarifies conceptual tensions in cosmology and unification.

4 Time as an Emergent Triadic Construct

Time occupies a peculiar position in modern physics. In quantum mechanics, time appears as an external parameter rather than an operator. In general relativity, time is part of the dynamical geometry and varies with gravitational potential. In thermodynamics, time is associated with irreversibility and entropy increase. These differing roles suggest that time is not a fundamental entity but an emergent phenomenon arising from deeper structural relations.

Within the Triadic Cycle, time emerges from the interactions between energy, geometry, and information. Each role contributes a distinct aspect of temporal behaviour, and only their combined structure yields the familiar notion of time experienced in physical systems.

4.1 Time from Energy: Change and Quantum Speed Limits

Energy governs the rate at which physical systems evolve. In quantum mechanics, the Hamiltonian generates time evolution through the Schrödinger equation, and quantum speed limits place fundamental bounds on how quickly a state can change Margolus and Levitin 1998. The Mandelstam–Tamm bound relates the variance of energy to the minimal time required for a state to evolve into an orthogonal one:

$$\Delta t \geq \frac{\pi \hbar}{2\Delta E}.$$

These results show that temporal progression is inseparable from energetic resources. Time, in this sense, quantifies the rate of informational change driven by energy.

4.2 Time from Geometry: Causal Structure and Relativity

General relativity reveals that time is inseparable from geometry. Proper time along a worldline depends on the spacetime metric, and gravitational time dilation demonstrates that different observers experience different temporal rates depending on curvature Einstein 1922. Light cones define which events can influence one another, establishing a causal ordering that underlies temporal structure.

Thus, geometry contributes the relational and causal scaffolding from which temporal order emerges.

4.3 Time from Information: Distinguishability and Relational Clocks

Information-theoretic approaches further support the emergent nature of time. In the Page–Wootters mechanism Page and Wootters 1983, time arises from correlations between subsystems rather than from an external parameter. A system functions as a clock only insofar as its states can be distinguished and correlated with those of another system. Similarly, thermodynamic irreversibility arises from coarse-graining and the growth of informational entropy Jaynes 1957.

In this view, time reflects the structure of informational distinctions and their evolution.

4.4 Black Holes and the Breakdown of Universal Time

Black holes provide the clearest demonstration that time is not fundamental. For an external observer, time appears to freeze at the event horizon due to infinite redshift. For an infalling observer, proper time proceeds normally. This discrepancy shows that time is not an absolute parameter but a context-dependent manifestation of geometry, energy, and information.

The black-hole regime therefore exposes the triadic origin of time: energetic concentration shapes geometry, geometry shapes causal structure, and causal structure shapes informational accessibility.

4.5 Triadic Synthesis: Time as Emergent

Taken together, these perspectives reveal that time is not a primitive ingredient of physical law. Instead, it emerges from the interplay of the triadic roles:

- Energy determines the rate of change of informational states.
- Geometry determines causal structure and proper time.
- Information determines distinguishability and relational ordering.

Thus, time is an emergent construct arising from the closed triadic cycle:

$$E \rightarrow G \rightarrow I \rightarrow E.$$

This perspective resolves several conceptual tensions between quantum mechanics and general relativity and provides a unified account of temporal behaviour across physical regimes.

5 Black Holes as Triadic Extremes

Black holes provide the clearest and most extreme manifestation of the Triadic Cycle. They represent physical systems in which energy, geometry, and information interact at their theoretical limits. In this regime, the triadic roles become sharply delineated, and the closed cycle $E \rightarrow G \rightarrow I \rightarrow E$ is made explicit in ways not visible in less extreme settings. This section examines how black holes instantiate each triadic transition and why they serve as a canonical test case for the ontology proposed in this paper.

5.1 Energy Determines Geometry

The defining feature of a black hole is the concentration of energy within a sufficiently small region of spacetime. According to the Einstein field equations,

$$G_{\mu\nu} = 8\pi G T_{\mu\nu},$$

the stress–energy tensor fully determines the curvature of spacetime. A sufficiently dense energetic configuration produces an event horizon, a geometric boundary beyond which causal influence cannot escape Misner, Thorne, and Wheeler 1973. Thus, black holes exemplify the transition:

$$E \rightarrow G,$$

where energy dictates the global and local geometric structure.

5.2 Geometry Bounds Information

The geometry of a black hole imposes strict limits on the amount of information that can be stored or accessed. The Bekenstein–Hawking entropy formula Bekenstein 1973; Hawking 1975,

$$S_{\text{BH}} = \frac{k_B c^3}{4G\hbar} A,$$

relates the entropy of a black hole to the area A of its event horizon, not its volume. This area law demonstrates that geometry directly constrains informational capacity. Furthermore, causal structure near the horizon determines which degrees of freedom are accessible to external observers.

Black holes therefore realize the transition:

$$G \rightarrow I,$$

where geometric constraints determine informational limits.

5.3 Information Constrains Energetic Evolution

The unitarity of quantum mechanics requires that information cannot be destroyed. This principle constrains the energetic evolution of black holes, most notably through Hawking radiation. If black holes were to irreversibly erase information, the resulting non-unitary evolution would violate fundamental principles of quantum theory Preskill 1992. The Page curve Page 1993 describes how information must be encoded in the radiation to preserve unitarity.

Thus, informational conservation shapes the energetic behaviour of black holes, realizing:

$$I \rightarrow E.$$

5.4 Black Holes as Computational Extremes

Black holes saturate several theoretical bounds on computation and information processing. They achieve the maximum entropy permitted by a given surface area and saturate the Bekenstein bound on information storage Bekenstein 1981. They also reach the fastest possible rate of quantum evolution allowed by energy constraints, approaching the Margolus–Levitin limit Margolus and Levitin 1998. In this sense, black holes function as optimal information processors within the laws of physics.

5.5 Breakdown of Universal Time

Black holes also reveal the emergent nature of time. For an external observer, time appears to slow and freeze at the event horizon due to infinite gravitational redshift. For an infalling observer, proper time proceeds normally. This discrepancy shows that time is not a universal parameter but a context-dependent manifestation of the triadic roles: energetic concentration shapes geometry, geometry shapes causal structure, and causal structure shapes informational accessibility.

5.6 Dark Matter and Dark Energy Fail the Black-Hole Test

The black-hole regime exposes a conceptual tension in the traditional classification of dark matter and dark energy. If dark matter were an energetic substance, it would contribute to black-hole entropy, microstates, and Hawking radiation. However, no such contributions are observed or theoretically required. Similarly, dark energy does not behave as an energetic quantity in black-hole thermodynamics; it appears only as a geometric term in the cosmological constant.

This suggests that both concepts are misclassified within the matter–energy vocabulary and are better understood as structural geometric roles rather than physical substances.

5.7 The Full Triadic Cycle in Black Holes

Black holes uniquely expose the entire triadic cycle:

- Energy determines geometry through gravitational collapse.
- Geometry bounds information through horizon area and causal structure.
- Information constrains energetic evolution through unitarity and Hawking radiation.

No other known physical system makes the closed sequence $E \rightarrow G \rightarrow I \rightarrow E$ so explicit. Black holes therefore serve as the canonical demonstration of the Triadic Cycle and provide strong evidence that the triadic roles are not imposed abstractions but structural features of physical law.

6 Cosmology Through the Triad

Cosmology provides a natural arena for examining the large-scale consequences of the Triadic Cycle. The evolution of the universe involves the interplay of energy distribution, geometric expansion, and informational structure across cosmological scales. While cosmology is often framed in terms of matter content, expansion rates, and metric evolution, the Triadic Cycle reveals a deeper architectural pattern: cosmic history can be understood as a recursive reorganization of the triadic roles.

6.1 Large-Scale Information Flow

The early universe was characterized by extremely low entropy and high uniformity, implying a highly constrained informational state Penrose 1979. As the universe expanded, information capacity increased through the growth of causal horizons and the formation of structure. Gravitational clustering, galaxy formation, and cosmic web dynamics can be interpreted as processes that redistribute and reorganize information under geometric and energetic constraints.

In this view, cosmological evolution reflects the transition:

$$G \rightarrow I,$$

where expanding geometry increases informational capacity and enables the emergence of complex structure.

6.2 Energy, Expansion, and Geometric Evolution

The Friedmann equations relate the expansion rate of the universe to its energy content:

$$H^2 = \frac{8\pi G}{3}\rho - \frac{k}{a^2} + \frac{\Lambda}{3}.$$

Here, energy density ρ and the cosmological constant Λ determine the large-scale geometry and expansion dynamics Weinberg 2008. This is a direct cosmological realization of the triadic transition:

$$E \rightarrow G.$$

Matter, radiation, and vacuum energy each shape the geometry in distinct ways, and the resulting curvature determines causal structure, horizon size, and the flow of information.

6.3 Holography and Cosmic Horizons

Cosmological horizons, such as the de Sitter horizon, impose limits on the amount of information accessible to any observer. The Gibbons–Hawking entropy of de Sitter space Gibbons and Hawking 1977 mirrors the Bekenstein–Hawking entropy of black holes:

$$S_{\text{dS}} = \frac{A}{4G\hbar}.$$

This suggests that cosmic geometry encodes information holographically, reinforcing the triadic transition:

$$G \rightarrow I.$$

The holographic interpretation of cosmology implies that large-scale geometry is not merely a background but an informational boundary condition.

6.4 Cosmic Evolution as Recursive Triadic Reorganization

The history of the universe can be interpreted as a sequence of triadic reorganizations:

- **Inflation:** Energetic vacuum drives rapid geometric expansion ($E \rightarrow G$), increasing informational capacity.
- **Recombination:** Geometry stabilizes, enabling the formation of distinguishable structures ($G \rightarrow I$).
- **Structure formation:** Information-rich configurations feed back into energetic and gravitational dynamics ($I \rightarrow E$).
- **Late-time acceleration:** Vacuum energy again shapes geometry ($E \rightarrow G$).

Cosmic history thus reflects repeated cycles of triadic interaction rather than a linear progression of independent physical processes.

6.5 Phase Transitions in Complexity

Cosmological epochs can be viewed as phase transitions in the complexity of the triadic roles. During inflation, the universe was dominated by a single energetic degree of freedom. During matter domination, geometry and information became increasingly structured. In the current dark-energy-dominated era, geometry again plays the leading role in determining causal structure and informational accessibility.

These transitions illustrate how the dominance of each triadic role shifts over cosmic time, while the underlying cycle remains intact.

6.6 Triadic Interpretation of Cosmology

The Triadic Cycle provides a coherent architectural interpretation of cosmology:

- Energy content determines large-scale geometry and expansion.
- Geometry determines causal horizons and informational capacity.
- Information-rich structure feeds back into energetic and gravitational dynamics.

Cosmology therefore exemplifies the recursive and large-scale manifestation of the triadic sequence:

$$E \rightarrow G \rightarrow I \rightarrow E.$$

This perspective does not modify the standard cosmological model but clarifies its conceptual structure and highlights the deep interdependence of energy, geometry, and information in the evolution of the universe.

7 Dark Matter and Dark Energy as Non-Compatible Roles

Dark matter and dark energy constitute the majority of the universe's inferred content, yet their physical nature remains unknown. Despite their names, neither behaves like conventional matter or energy. Within the Triadic Cycle, this mismatch becomes conceptually clear: both phenomena fail to occupy the energetic or informational roles of the Triad and are more coherently interpreted as structural geometric constraints rather than physical substances. This section examines the ontological tension between their observed behaviour and the traditional matter–energy vocabulary.

7.1 The Problem of Misleading Nomenclature

The terms “dark matter” and “dark energy” implicitly suggest that these components belong to the energetic sector of physical ontology. However, this classification is based on historical analogy rather than structural behaviour. Dark matter does not interact electromagnetically, thermodynamically, or informationally, and dark energy does not behave as an energetic quantity capable of performing work or contributing to entropy. Their naming therefore obscures their actual roles within physical theory.

7.2 Dark Matter: A Geometric Role Misclassified as Substance

Dark matter is inferred from gravitational effects: galaxy rotation curves, gravitational lensing, and large-scale structure formation Rubin and Ford 1970; Clowe et al. 2006. Yet no non-gravitational interaction has been observed. Dark matter:

- does not emit, absorb, or scatter light,
- does not thermalize or contribute to entropy,
- does not participate in known particle interactions,
- does not exhibit microstates accessible to observers.

Within the Triadic Cycle, these properties indicate that dark matter does not occupy the energetic role E or the informational role I . Its behaviour is purely geometric: it modifies curvature without carrying observable energy in the conventional sense. Thus, dark matter is more coherently interpreted as a structural geometric constraint rather than a physical substance.

7.3 Dark Energy: Geometry Without Energetics

Dark energy is inferred from the accelerated expansion of the universe Riess et al. 1998; Perlmutter et al. 1999. In the standard model, it is represented by a cosmological constant Λ , which contributes a uniform curvature term to the Einstein equations. However, dark energy:

- has no temperature or entropy,
- does not perform work in the thermodynamic sense,
- does not fluctuate or cluster,
- does not correspond to excitations of any known field.

These properties are incompatible with the energetic role of the Triad. Instead, dark energy behaves as a purely geometric term, altering large-scale curvature without participating in informational or energetic dynamics. Its classification as “energy” is therefore ontologically misleading.

7.4 The Black-Hole Consistency Test

Black holes provide a stringent test for the ontological status of dark matter and dark energy. If dark matter were an energetic substance, it would contribute to black-hole entropy, microstates, and Hawking radiation. No such contributions are predicted or observed. Similarly, dark energy does not appear in black-hole thermodynamics except through its effect on asymptotic geometry; it does not modify horizon microstates or entropy.

This absence is consistent with the interpretation that both dark components are geometric constraints rather than energetic entities. The black-hole regime therefore exposes the incompatibility of their traditional classification with the triadic roles.

7.5 Triadic Interpretation of Dark Components

Within the Triadic Cycle, the behaviour of dark matter and dark energy aligns naturally with the geometric role:

- Dark matter modifies curvature without carrying accessible energy or information.
- Dark energy modifies large-scale curvature without thermodynamic or informational properties.

Neither phenomenon fits into the energetic role E or the informational role I . Their observed behaviour is fully compatible with the geometric role G and incompatible with the traditional matter–energy vocabulary.

7.6 Conceptual Implications

This reframing does not introduce new physics or propose alternative models. Instead, it clarifies an ontological mismatch: the dark components behave as geometric constraints, not as substances. Recognizing this resolves several conceptual tensions in cosmology and aligns the dark sector with the structural ontology of the Triadic Cycle.

The Triad therefore provides a coherent and non-speculative explanation for why dark matter and dark energy have resisted particle-based or energetic interpretations: they were never energetic entities to begin with.

8 Mainstream Theories as Architectural Projections

The major theoretical frameworks of modern physics are often viewed as competing or mutually incompatible descriptions of reality. Quantum mechanics, quantum field theory, general relativity, thermodynamics, and holography each emphasize different structural aspects of the physical world, leading to conceptual tensions that have motivated decades of unification attempts. Within the Triadic Cycle, these tensions acquire a natural explanation: each theory can be understood as a *projection* of the full triadic structure onto one of its fundamental roles.

This architectural interpretation does not replace existing theories or alter their empirical content. Instead, it clarifies why each theory succeeds within its domain, why they appear incompatible when naively combined, and how they collectively instantiate the full triadic ontology.

8.1 Quantum Mechanics as an Information-First Projection

Quantum mechanics is fundamentally concerned with informational structure. States are represented by vectors in a Hilbert space, and physical evolution is governed by unitary transformations that preserve informational distinguishability Nielsen and Chuang 2010. Measurement, entanglement, and decoherence are informational processes, and the theory’s mathematical formalism is optimized for describing correlations and state spaces rather than geometric or energetic structure.

From the triadic perspective, quantum mechanics is a projection onto the informational role:

$$\text{QM} \longrightarrow I.$$

This explains both its strengths (precision in describing correlations and state evolution) and its limitations (lack of intrinsic geometry or dynamical spacetime).

8.2 Quantum Field Theory as an Energy-First Projection

Quantum field theory (QFT) emphasizes energetic and dynamical structure. Fields carry excitations whose energies determine particle content, interactions, and propagation. Renormalization, vacuum structure, and symmetry breaking are energetic phenomena Peskin and Schroeder 1995. Geometry enters only indirectly through locality and Lorentz invariance.

Thus, QFT is a projection onto the energetic role:

$$\text{QFT} \longrightarrow E.$$

This explains why QFT excels at describing interactions and dynamics but does not provide a dynamical spacetime or a geometric account of gravity.

8.3 General Relativity as a Geometry-First Projection

General relativity (GR) is a geometry-first theory. Spacetime curvature encodes gravitational dynamics, and causal structure is determined by the metric Einstein 1922. Energy and momentum influence geometry through the Einstein field equations, but the theory's primary object is the geometric role.

Thus, GR is a projection onto the geometric role:

$$\text{GR} \longrightarrow G.$$

This explains why GR provides a coherent account of spacetime and gravity but does not incorporate quantum informational structure or energetic microstates.

8.4 Thermodynamics as an Information–Energy Projection

Thermodynamics and statistical mechanics occupy a hybrid position between information and energy. Entropy, free energy, and the second law link informational structure to energetic behaviour Jaynes 1957. Landauer's principle Landauer 1961 explicitly connects information erasure to energetic cost.

Thus, thermodynamics can be viewed as a projection onto the $I \rightarrow E$ transition:

$$\text{Thermodynamics} \longrightarrow (I \rightarrow E).$$

8.5 Holography as a Geometry–Information Projection

The holographic principle and the AdS/CFT correspondence reveal a deep connection between geometry and information. The Ryu–Takayanagi formula Ryu and Takayanagi 2006 relates entanglement entropy to geometric surfaces, and tensor-network models show how geometry can emerge from entanglement structure Swingle 2012.

Thus, holography is a projection onto the $G \rightarrow I$ transition:

$$\text{Holography} \longrightarrow (G \rightarrow I).$$

8.6 Why Theories Appear Incompatible

The apparent incompatibilities between mainstream theories arise from the fact that each emphasizes a different triadic role:

- QM emphasizes information and lacks dynamical geometry.
- QFT emphasizes energy and lacks dynamical spacetime.
- GR emphasizes geometry and lacks quantum informational structure.

These differences are not contradictions but architectural projections. Each theory captures a partial view of the full triadic structure, optimized for its domain of applicability.

8.7 Architectural Unification Without Replacement

The Triadic Cycle provides a unifying ontology without requiring new physics or modifying existing theories. It clarifies:

- why each theory succeeds in its domain,
- why they appear incompatible when naively combined,
- and how they collectively instantiate the full triadic cycle.

This architectural perspective does not constitute a theory of everything, but it provides a minimal ontological framework that unification efforts can build upon and remain consistent with. It offers a structural explanation for the fragmentation of modern physics and a coherent foundation for conceptual integration.

9 Related Work

The Triadic Cycle intersects with several major research traditions in contemporary theoretical physics, including information-theoretic foundations, emergent spacetime, quantum gravity, thermodynamics, and cosmological interpretation. This section situates the Triadic Cycle within these domains while emphasizing that it does not belong exclusively to any of them. Instead, it provides an architectural ontology that clarifies their interrelations.

9.1 Information-Theoretic Foundations of Physics

Information has increasingly been recognized as a central concept in physics. Quantum information theory formalizes the structure of quantum states, entanglement, and unitary evolution Nielsen and Chuang 2010; Wilde 2017. The role of information in thermodynamics is highlighted by Jaynes’ maximum-entropy formalism Jaynes 1957, Landauer’s principle Landauer 1961, and the work of Bennett on reversible computation Bennett 1973. The connection between information and spacetime has been explored in the context of entanglement entropy Bombelli, Koul, et al. 1986; Srednicki 1993 and quantum error correction Almheiri, Dong, and Harlow 2015.

The Triadic Cycle aligns with these developments by treating information as a fundamental ontological role, while extending them by embedding information within a closed cycle that also includes energy and geometry.

9.2 Geometry and Emergent Spacetime

A large body of work investigates the emergence of spacetime geometry from more primitive informational or algebraic structures. The holographic principle ’t Hooft 1993; Susskind 1995 and the AdS/CFT correspondence Maldacena 1998 provide explicit realizations of geometry–information duality. The Ryu–Takayanagi formula Ryu and Takayanagi 2006 and its quantum corrections Faulkner, Lewkowycz, and Maldacena 2013 relate entanglement entropy to geometric surfaces. Tensor-network models such as MERA Vidal 2007 and the work of Swingle Swingle 2012 suggest that geometry may emerge from entanglement structure.

The Triadic Cycle incorporates these insights but does not reduce geometry to information; instead, geometry is treated as a distinct ontological role that both constrains and is constrained by information and energy.

9.3 Energy, Dynamics, and Quantum Field Theory

Quantum field theory (QFT) provides the dominant framework for describing energetic and dynamical processes Peskin and Schroeder 1995; Weinberg 1995. Renormalization group flows Wilson 1975, vacuum structure, and symmetry breaking Nambu 2009 are energetic phenomena. The stress–energy tensor plays a central role in both QFT and general relativity, linking energetic content to geometric curvature.

The Triadic Cycle captures this by recognizing energy as a fundamental role and situating QFT as a projection onto the energetic sector of the triadic ontology.

9.4 Quantum Gravity and Unification Attempts

Numerous approaches attempt to reconcile quantum mechanics and general relativity. String theory Polchinski 1998, loop quantum gravity Rovelli 2004, causal sets Bombelli, Lee, et al. 1987, asymptotic safety Weinberg 1979, and non-commutative geometry Connes 1994 each introduce new mathematical structures or microphysical assumptions. These frameworks typically aim to unify geometry and quantum information at a fundamental level.

The Triadic Cycle differs from these approaches by not proposing new entities or dynamics. Instead, it identifies a minimal ontological structure already implicit in existing theories, making it compatible with a wide range of unification programs while remaining agnostic about their specific mechanisms.

9.5 Thermodynamics, Entropy, and Irreversibility

The deep connections between information, entropy, and energy have been explored in statistical mechanics Callen 1985, black-hole thermodynamics Bekenstein 1973; Hawking 1975, and quantum thermodynamics Goold et al. 2016. The second law has been reinterpreted in informational terms Parrondo, Horowitz, and Sagawa 2015, and quantum speed limits Margolus and Levitin 1998; Deffner and Campbell 2017 link energetic resources to informational change.

The Triadic Cycle incorporates these results by treating the $I \rightarrow E$ transition as a fundamental structural relation.

9.6 Cosmology and the Dark Sector

The nature of dark matter and dark energy remains one of the central open questions in cosmology. Observational evidence for dark matter includes galaxy rotation curves Rubin and Ford 1970, gravitational lensing Clowe et al. 2006, and cosmic microwave background anisotropies Collaboration 2020. Dark energy is inferred from the accelerated expansion of the universe Riess et al. 1998; Perlmutter et al. 1999 and is commonly modeled via a cosmological constant Weinberg 2008.

Alternative interpretations include modified gravity Milgrom 1983, emergent gravity Verlinde 2017, and geometric approaches to cosmic acceleration Clifton et al. 2012. The Triadic Cycle contributes to this discussion by offering an ontological explanation for why dark matter and dark energy behave anomalously: their properties align with the geometric role rather than the energetic or informational roles.

9.7 Conceptual Foundations and Structural Ontologies

Several authors have explored structural or relational ontologies in physics, including structural realism Ladyman 1998, causal set theory Bombelli, Lee, et al. 1987, and relational interpretations of quantum mechanics Rovelli 1996. The Triadic Cycle shares the structural emphasis of these approaches but differs by identifying a minimal closed set of roles—information, energy, and geometry—and articulating their mutual determination.

9.8 Positioning of the Triadic Cycle

The Triadic Cycle is situated at the intersection of:

- information-theoretic physics,
- emergent spacetime and holography,
- energetic and dynamical frameworks in QFT,
- quantum gravity and unification attempts,
- thermodynamics and entropy,
- and cosmological interpretation of the dark sector.

It does not compete with these approaches but provides a minimal ontological structure that clarifies their interrelations. By identifying information, energy, and geometry as the fundamental roles of physical ontology and by articulating their closed cycle of mutual determination, the Triadic Cycle offers a unifying architectural perspective that complements and contextualizes existing work.

10 Testable Predictions and Conceptual Consequences

The Triadic Cycle is an ontological framework rather than a dynamical theory, and it does not introduce new physical entities or equations. Nevertheless, the framework yields several conceptual consequences and observational discriminators that can guide empirical investigation. These consequences arise from the structural constraints imposed by the triadic roles and from the requirement that physical phenomena must be compatible with the closed cycle

$$E \rightarrow G \rightarrow I \rightarrow E.$$

This section outlines testable implications and conceptual predictions that follow from the Triadic Cycle without invoking speculative microphysics.

10.1 Prediction 1: Dark Components Should Exhibit Purely Geometric Behaviour

If dark matter and dark energy occupy the geometric role rather than the energetic or informational roles, then:

- they should not contribute to entropy or microstate counting,
- they should not participate in thermodynamic processes,
- they should not appear in Hawking radiation spectra,
- they should not exhibit particle-like interactions or decay channels,
- and they should influence dynamics only through curvature.

These criteria provide observational discriminators between geometric and particle-based interpretations of the dark sector. Future precision measurements of black-hole thermodynamics, gravitational lensing, and cosmic expansion can test these predictions.

10.2 Prediction 2: Horizon Entropy Should Remain Area-Law Dominated

The Triadic Cycle implies that informational capacity is fundamentally geometric. Thus, horizon entropy—whether for black holes, de Sitter horizons, or causal diamonds—should continue to obey area-law scaling Bekenstein 1973; Hawking 1975; Gibbons and Hawking 1977. Any deviation from area-law behaviour would indicate a breakdown of the $G \rightarrow I$ transition and challenge the triadic ontology.

This prediction is testable through:

- quantum-gravity corrections to black-hole entropy,
- entanglement entropy in holographic systems,
- and numerical simulations of quantum extremal surfaces.

10.3 Prediction 3: Information Conservation Should Constrain High-Energy Dynamics

The Triadic Cycle requires that information constrains energetic evolution ($I \rightarrow E$). Thus, any high-energy process—including black-hole evaporation, early-universe dynamics, or quantum-gravity scattering—must preserve informational structure.

Observable consequences include:

- Page-curve-consistent evaporation of black holes,
- absence of non-unitary evolution in quantum-gravity regimes,
- and consistency between entanglement structure and energetic flow.

These predictions align with recent developments in quantum gravity and holography Almheiri, Dong, and Harlow 2015.

10.4 Prediction 4: Geometry Should Encode Information in All Causal Horizons

The triadic transition $G \rightarrow I$ implies that any causal horizon—not only black-hole horizons—should encode information holographically. This includes:

- Rindler horizons,
- cosmological horizons,
- apparent horizons in dynamical spacetimes,
- and entanglement wedges in holographic dualities.

This prediction is testable through studies of entanglement entropy, quantum extremal surfaces, and horizon thermodynamics.

10.5 Prediction 5: No Physical System Should Exhibit Purely Energetic or Informational Behaviour

Because the triadic roles form a closed cycle, no physical system should instantiate only one role in isolation. For example:

- purely energetic models without geometric backreaction should be incomplete,

- purely informational models without energetic cost should violate quantum speed limits,
- and purely geometric models without informational bounds should violate holographic constraints.

This provides a structural criterion for evaluating theoretical proposals in quantum gravity, cosmology, and emergent spacetime.

10.6 Conceptual Consequence: Unification Without New Entities

The Triadic Cycle suggests that unification may not require new particles, fields, or micro-physical structures. Instead, unification may emerge from recognizing that existing theories are projections of a single architectural ontology. This perspective predicts that:

- future unification frameworks should integrate the triadic roles coherently,
- apparent incompatibilities between theories should reflect projection differences,
- and successful unification should preserve the closed triadic cycle.

This is a conceptual rather than empirical prediction, but it provides a guiding principle for theoretical development.

10.7 Summary of Testable Implications

Although the Triadic Cycle does not introduce new dynamics, it yields a set of structural predictions:

- dark components should behave geometrically, not energetically;
- horizon entropy should remain area-law dominated;
- information conservation should constrain high-energy processes;
- geometry should encode information across all causal horizons;
- and no physical system should instantiate a single role in isolation.

These predictions provide observational and conceptual discriminators that can be tested through cosmology, black-hole physics, holography, and quantum information.

11 Limitations, Open Questions, Novelties, and Implications

The Triadic Cycle is proposed as a minimal ontological framework extracted from structural patterns already present in established physics. As such, it offers conceptual clarity rather than new dynamical laws. This section outlines the limitations of the framework, identifies open questions for future work, highlights its conceptual novelties, and discusses broader implications for theoretical physics.

11.1 Limitations

The Triadic Cycle has several inherent limitations:

- **No new dynamics.** The framework does not introduce new equations of motion or microphysical mechanisms. It therefore cannot make quantitative predictions beyond those already implied by existing theories.
- **No commitment to specific quantum-gravity models.** The Triad is compatible with multiple approaches to quantum gravity (string theory, loop quantum gravity, holography, causal sets), but it does not select among them.
- **No microscopic account of geometry.** While the Triad clarifies the ontological role of geometry, it does not specify the microstructure of spacetime or the degrees of freedom underlying curvature.
- **No direct explanation of cosmological parameters.** The framework does not predict the numerical values of the cosmological constant, dark-matter distribution, or inflationary parameters.
- **Conceptual rather than computational.** The Triad provides an architectural perspective but does not yet yield computational tools or calculational techniques.

These limitations reflect the framework’s purpose: it is an ontological extraction, not a replacement for existing physical theories.

11.2 Open Questions

The Triadic Cycle raises several questions that merit further investigation:

- **Can the triadic roles be formalized mathematically?** A rigorous algebraic or categorical formulation of the Triad may clarify its structural properties.
- **What is the microphysical origin of the triadic transitions?** Are the transitions $E \rightarrow G$, $G \rightarrow I$, and $I \rightarrow E$ emergent from deeper principles, or are they primitive?
- **How does the Triad interface with quantum gravity?** Can the triadic roles be mapped onto the degrees of freedom in specific quantum-gravity proposals?
- **Is there a triadic analogue of Noether’s theorem?** If the roles correspond to conserved structural quantities, what symmetries underlie these conservation laws?
- **Can cosmological evolution be expressed as a triadic dynamical system?** A formal triadic description of cosmic epochs may reveal new structural regularities.
- **How universal is the triadic ontology?** Does it apply only to our universe, or is it a structural feature of any physically coherent theory?

These questions point toward future research directions that could deepen the framework or connect it to existing mathematical structures.

11.3 Conceptual Novelties

The Triadic Cycle introduces several conceptual innovations:

- **Minimal closed ontology.** The Triad identifies information, energy, and geometry as the minimal set of roles required for physical coherence.
- **Architectural unification.** The framework unifies mainstream theories not by extending them but by interpreting them as projections of a single ontological structure.
- **Reclassification of the dark sector.** The Triad provides a non-speculative explanation for the anomalous behaviour of dark matter and dark energy by interpreting them as geometric roles.
- **Emergent time from triadic interaction.** The framework offers a unified explanation for the differing roles of time in QM, GR, and thermodynamics.
- **Black holes as triadic extremal systems.** The Triad clarifies why black holes uniquely expose the full cycle of mutual determination.

These novelties are conceptual rather than dynamical, but they offer a coherent reinterpretation of foundational physics.

11.4 Implications for Future Research

The Triadic Cycle has several implications for theoretical physics:

- **Unification efforts.** Any future unification framework should integrate the triadic roles coherently and remain compatible with the closed cycle.
- **Interpretation of quantum gravity.** The Triad may serve as a guiding ontology for evaluating quantum-gravity proposals based on their treatment of information, energy, and geometry.
- **Cosmology.** The triadic interpretation of dark components suggests new ways to analyze cosmological data without invoking speculative particles.
- **Foundations of thermodynamics.** The $I \rightarrow E$ transition provides a structural explanation for the energetic cost of information processing.
- **Black-hole physics.** The Triad predicts that horizon thermodynamics should remain area-law dominated and that information conservation should constrain high-energy processes.

These implications highlight the potential of the Triadic Cycle to inform both conceptual and empirical research.

11.5 Summary

The Triadic Cycle is a minimal, closed ontological framework that clarifies the structural relationships between information, energy, and geometry. While it does not introduce new dynamics, it offers conceptual novelties, raises meaningful open questions, and provides testable implications that can guide future theoretical and observational work.

12 Conclusion

This paper has introduced the Triadic Cycle as a minimal and closed ontological framework consisting of three fundamental roles—information, energy, and geometry—and has shown that this structure is already implicit across the major theoretical frameworks of modern physics. Rather than proposing new entities or dynamical laws, the Triadic Cycle offers an architectural extraction of the structural commitments shared by quantum mechanics, quantum field theory, general relativity, thermodynamics, and holography. Each of these theories can be understood as a projection onto one or more triadic roles, and their apparent incompatibilities arise from differences in projection rather than from contradictions in physical content.

Black holes provide the clearest and most extreme realization of the triadic structure. In these systems, energy determines geometry, geometry bounds information, and information constrains energetic evolution. This regime also reveals the emergent nature of time and exposes the conceptual mismatch in the traditional classification of dark matter and dark energy as energetic substances. Interpreting the dark sector as geometric rather than energetic aligns with observational behaviour while avoiding speculative microphysical assumptions.

The Triadic Cycle yields several conceptual consequences and observational discriminators. It predicts that horizon entropy should remain area-law dominated, that information conservation should constrain high-energy processes, and that dark components should exhibit purely geometric behaviour. These implications do not rely on new physics but follow from the structural coherence of the triadic roles.

By identifying information, energy, and geometry as the minimal ontological components of physical theory and by articulating their closed cycle of mutual determination, the Triadic Cycle provides a unifying architectural perspective that clarifies the relationships between established theories. It offers a conceptual foundation upon which future unification efforts can build and remain consistent, while raising meaningful open questions about the mathematical formalization of the triadic roles, their microphysical origins, and their implications for quantum gravity and cosmology.

The Triadic Cycle does not claim to be a final theory or a replacement for existing frameworks. Instead, it aims to illuminate the structural architecture underlying modern physics and to provide a coherent ontological basis for interpreting its most successful theories. In doing so, it offers a pathway toward conceptual unification that is grounded not in speculative extensions but in the shared foundations of the physics we already trust.

A Glossary of Triadic Roles

Information (I). The role responsible for distinction, state, and representational capacity. Information determines what can be distinguished, stored, correlated, or transformed. In physics, it appears through Hilbert-space structure, entropy, entanglement, and coarse-graining. In the Triadic Cycle, information constrains energetic evolution and is bounded by geometric structure.

Energy (E). The role responsible for transformation, dynamics, and change. Energy generates time evolution in quantum mechanics, drives interactions in quantum field theory, and determines curvature through the stress–energy tensor in general relativity. In the Triadic Cycle, energy shapes geometry and is constrained by informational structure.

Geometry (G). The role responsible for relation, locality, and causal structure. Geometry determines which interactions are possible, how information can propagate, and how energetic content shapes spacetime. In the Triadic Cycle, geometry bounds informational capacity and is generated by energetic configurations.

The Triadic Cycle. A minimal closed sequence of mutual determination:

$$E \rightarrow G \rightarrow I \rightarrow E.$$

Energy shapes geometry; geometry bounds information; information constrains energetic evolution. This cycle is not imposed but extracted from established physics, appearing in relativity, thermodynamics, quantum information, and black-hole thermodynamics.

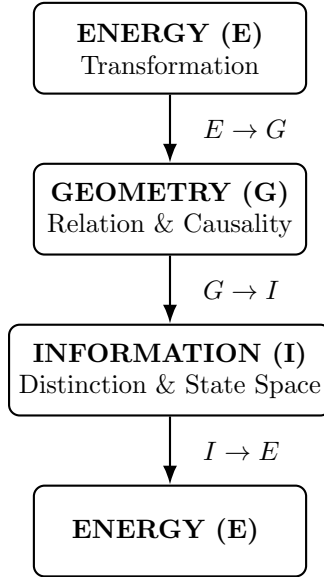
Minimality. The triad is minimal because:

- mass is not a separate primitive (being equivalent to energy),
- geometry is derived rather than fundamental,
- information is physically instantiated rather than abstract.

Removing any role breaks the coherence of physical description.

Recursion. Each role generates the next, forming a recursive engine that appears across scales: from black holes to quantum information, thermodynamic computation, and cosmological evolution. This recurrence suggests that the triad is a structural feature of physical law rather than a modelling artefact.

Triadic Cycle Diagram.



This diagram summarizes the closed sequence of mutual determination:

$$E \rightarrow G \rightarrow I \rightarrow E.$$

Energy shapes geometry; geometry bounds information; information constrains energetic evolution. The cycle is minimal, closed, and recursive, appearing across quantum mechanics, general relativity, thermodynamics, holography, and black-hole physics.

B Triadic Cosmos Ecosystem Context

This paper is part of the *Triadic Cosmos* ecosystem, which develops a unified conceptual and computational framework spanning physical ontology and modular architectures for intelligent systems. All materials, extended notes, and evolving documentation are available in the public repository: <https://github.com/triadic-cosmos>.

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